# SINGAPORE SCHOOL KINDERLAND

**Department of Information Technology & Computer Science · Academic Year 2026–2027**

**CAMBRIDGE O LEVEL COMPUTER SCIENCE (0478/2210) · CLASS TEST 1 (TERM 2)**

Topic 2.1: Types and Methods of Data Transmission | Total: 25 Marks | Time: 45 Mins

| | | |
|---|---|---|
| **Candidate Name:** _____ | **Centre / Roll No:** _____ | **Class:** O Level [ ] |
| **Date of Examination:** _____ | **Duration:** 45 Mins | **Marks Awarded:** / 25 |

## SECTION A: TRANSMISSION MECHANICS & MULTIPLE CHOICE [5 MARKS]

1. Which transmission mode permits data to travel in both directions simultaneously across a communication medium?
(a) Simplex (b) Half-duplex (c) Full-duplex (d) Multiplex

2. Which transmission method transmits data one bit at a time over a single wire or optical fiber channel?
(a) Parallel (b) Serial (c) Synchronous (d) Asynchronous

3. What major transmission problem occurs in parallel transmission where bits arrive out of synchronization over long distances?
(a) Packet loss (b) Bit skew (c) Crosstalk (d) Attenuation

4. In packet switching, which component of a network packet contains the sender and destination IP addresses?
(a) Payload (b) Packet Trailer (c) Packet Header (d) Parity Block

5. Which interface standard uses serial transmission and four shielded wires (VBUS, GND, D+, D-) to connect peripherals?
(a) HDMI (b) VGA (c) USB (d) SATA

## SECTION B: SHORT STRUCTURED TECHNICAL QUESTIONS [6 MARKS]

1. **Packet Structure:** Identify the THREE distinct parts that make up a standard data packet: [3 Marks]

• Part 1: _____

• Part 2: _____

• Part 3: _____

2. **Packet Switching Mechanics:** Explain why consecutive packets belonging to the same document might travel along different physical routes across the Internet, and how the receiving computer reassembles them: [3 Marks]

Routing reason: _____

Reassembly method: _____

## ~~SECTION C: DIRECTIONAL TRANSMISSION CLASSIFICATION [4 MARKS]~~

For each communication scenario below, state whether it uses **Simplex**, **Half-duplex**, or **Full-duplex** transmission:

| Communication Scenario | Transmission Mode |
|---|---|
| A wireless computer mouse transmitting click coordinates to a USB dongle | _____ |
| A live two-way video conversation between doctors using Zoom / Microsoft Teams | _____ |
| Two security guards communicating via walkie-talkies (press-to-talk button) | _____ |
| A digital television station broadcasting a live football match to household antennas | _____ |

## SECTION D: SERIAL VS PARALLEL TRANSMISSION ANALYSIS [5 MARKS]

1. Contrast **Serial Transmission** and **Parallel Transmission** by completing the comparison table: [3 Marks]

| Comparison Feature | Serial Transmission | Parallel Transmission |
|---|---|---|
| Number of data transmission lines | _____ | _____ |
| Risk of bit skew / crosstalk | _____ | _____ |
| Typical operational distance | _____ | _____ |

2. Modern computers connect internal storage (SATA/NVMe) and external devices (USB) using high-speed serial connections rather than parallel cables (such as legacy IDE/PATA). State TWO technical reasons why serial has replaced parallel: [2 Marks]

Reason 1: .....

Reason 2: .....

## SECTION E: UNIVERSAL SERIAL BUS (USB) ARCHITECTURE [5 MARKS]

1. Outline THREE distinct benefits of using the **USB** connection system for connecting peripheral devices to a computer: [3 Marks]

Benefit 1: .....

Benefit 2: .....

Benefit 3: .....

2. State ONE limitation or drawback of the USB connection interface: [1 Mark]

.....

3. How does the host computer recognize a newly inserted USB device automatically without restarting (Plug-and-Play)? [1 Mark]

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